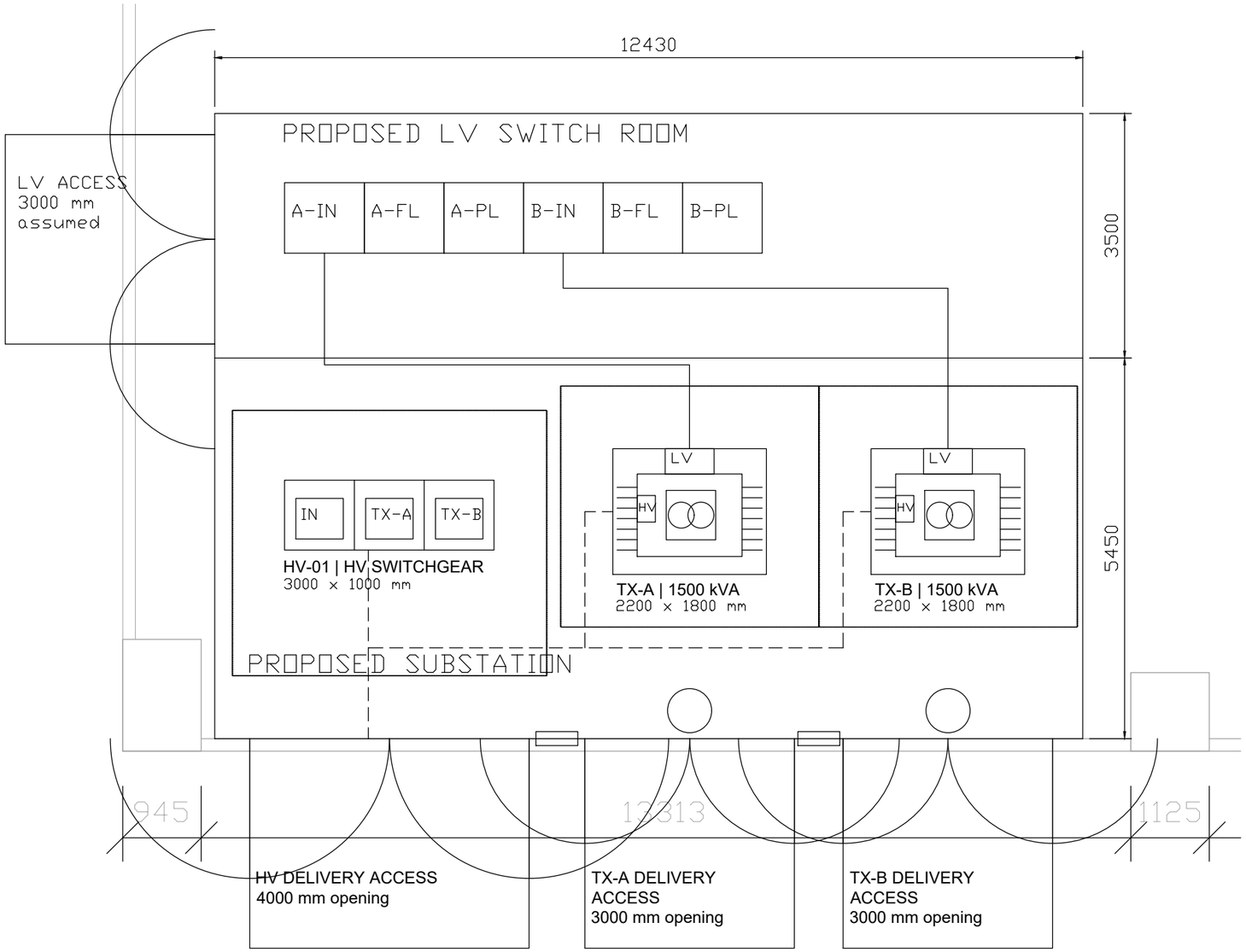
TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed = HV trench; solid = LV route; chain = clearance. Equipment details are diagrammatic.

LV INTERFACES / HV ROUTES
LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.
HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

CAPACITY AND EQUIPMENT
TX-A / TX-B: 1500 kVA each. Total 3000 kVA; design load 2672.216 kVA. No full-load N-1.
HV-01: 3000 x 1000 mm; TX-A/B: 2200 x 1800 mm.
Cabinets: A/B-IN incomer; A/B-FL floor feeders; A/B-PL plant/public feeders.
Generic plan symbols; overall dimensions are adopted design assumptions. Internal details are diagrammatic.

CLEARANCES / BUILDING WORKS
HV rear 1000, front 1800, sides 750. TX LV terminals north: 900; other sides 750.
Proposed 2h FRR partitions; fire-seal penetrations. FFL +0.15 vs external +0.00 assumed.
Actual structures, wall thickness and equipment dimensions require coordination.

ACCESS AND DOORS
South external access assumed. Provide 3000-wide clear delivery route; unloading distance <=20 m to verify.
Plant doors: HV 4000 W; TX 3000 W each; 2800 H assumed. Outward opening, shown at 90 degrees.
Adjacent plant doors used sequentially; TX-A has wicket personnel access.
Independent LV entrance west.

VENTILATION
2 low-level 600 W x 2000H inlet louvres: 2.4m2 gross x assumed 50% free area = 1.20m2 >=1.12m2. Certified free area >=46.67% required.
2 high-level 630-dia fan symbols are conditional references. Actual losses, fan duty and noise to confirm; outlet >=2.5m above street, away from people.

Basis: supplied COP215 Rev07; COP101 v16.
See the new main distribution schematic for TX-A/B, MSB-A/B and floor supply zones.